
UNSUPERVISED MACHINE LEARNING MODEL FOR CHARACTERIZING PHISHING ATTACKS

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PROBLEM



THE RISE OF AI HAS CONTRIBUTED TO THE DEVELOPMENT OF DIGITAL THREATS.



AV-ATLAS STATES THAT FOR AUGUST OF 2025 APPROXIMATELY 400 PHISHING EMAILS ARE DISCOVERED EACH DAY.



FOR THE FIRST QUARTER OF 2024 THE TOP 3 MOST PHISHED BRANDS WERE MICROSOFT, GOOGLE AND LINKEDIN



MICROSOFT RELATED CONTENT WAS DECLARED AS FAKE 40% OF THE TIMES.



THE SOLUTION

- Machine Learning model that helps to characterize Phishing emails.
- Interface for inputting email data and receive comments for which phishing Group does the email belong to.

PRELIMINARY PROCESS



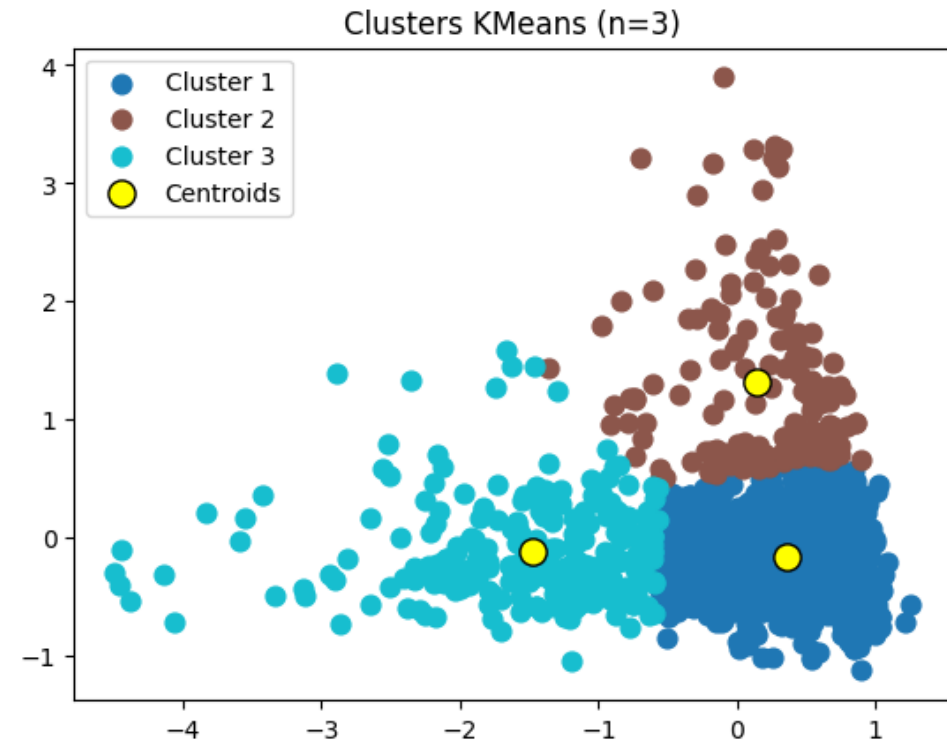
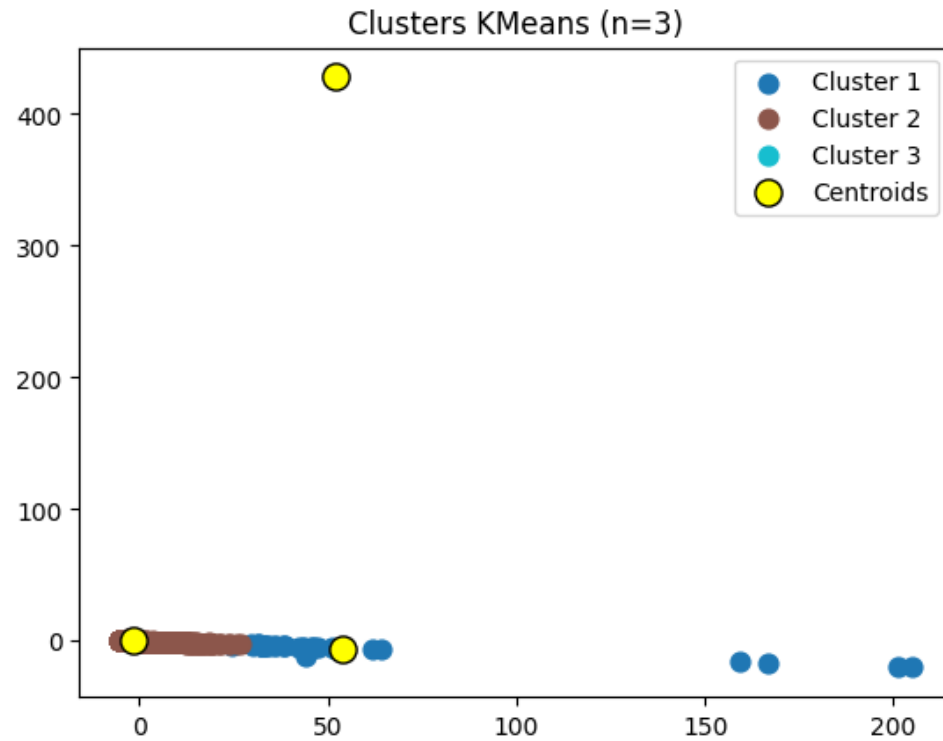
Data Recollection



Algorithm Selection

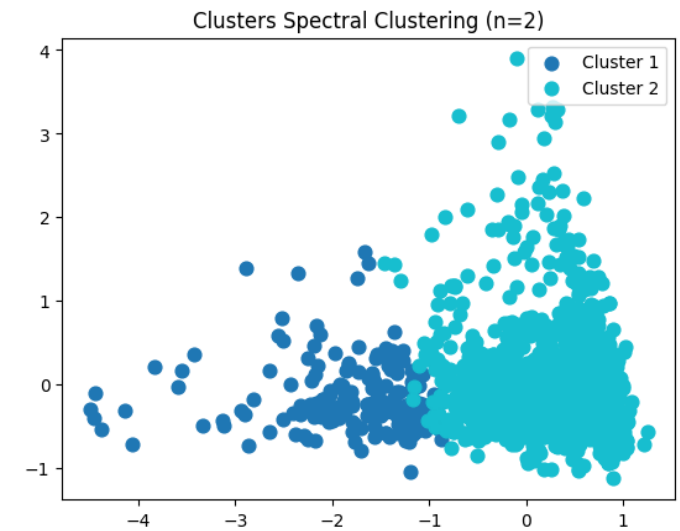
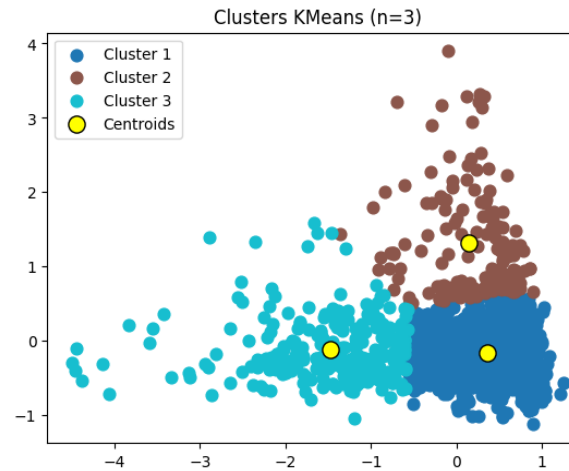
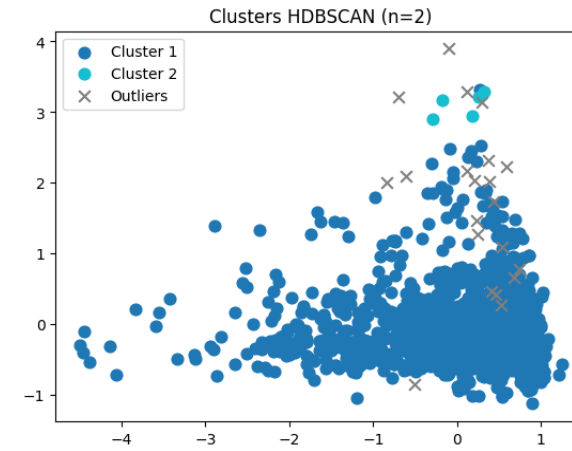
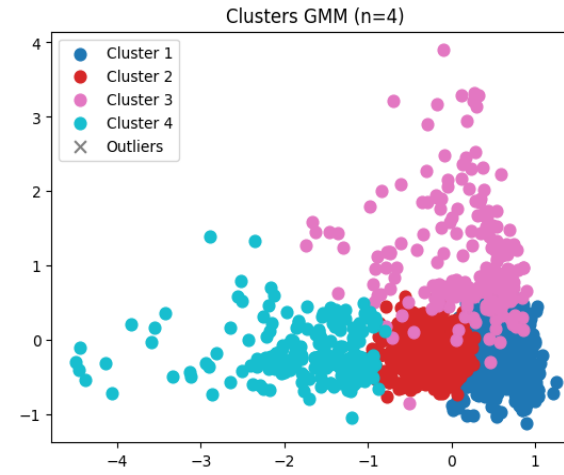
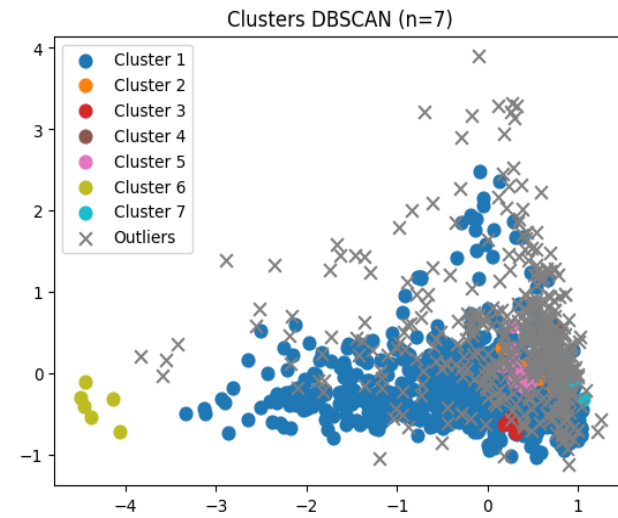


Text Technique Selection



PRELIMINARY PROCESS

PRELIMINARY PROCESS



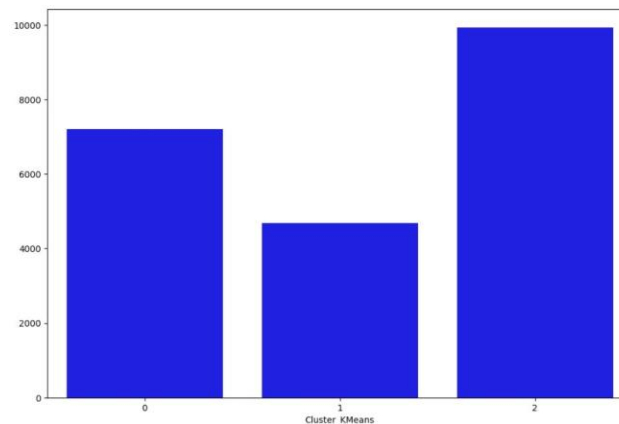
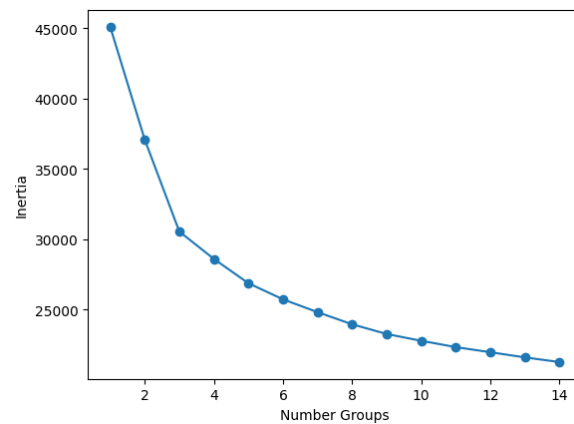
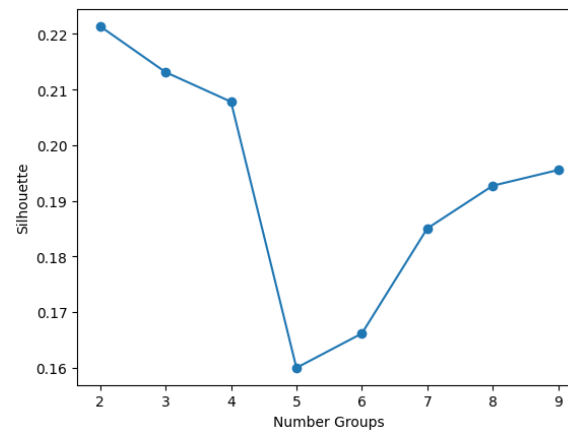
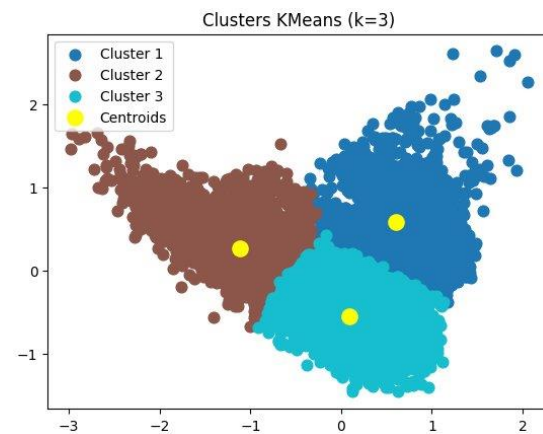
FINAL PROCESS



ALGORITHM SELECTED:
K-MEANS



TEXT TECHNIQUE
SELECTED: EMBEDDINGS



RESULTS

RESULTS



Cluster 0 -
News Alerts &
Social Content

Cluster 1 -
Replica Goods

Cluster 2 -
Pharmaceuticals
& Dating

INTERFACE



LIMITATIONS

1

Data may be assigned to wrong clusters

2

K-Means Inertia and Silhouette can achieve better scores

3

Training data may not be sufficient

CONCLUSIONS

Model allows to understand email structure

Model is not biased as no group has more than 50% of the training data

Phishing emails on their majority contain an URL

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